

CE 310 — Week 12 Weekly Quiz

Week: 12 | **Topic:** Sensitivity Analysis

Due: Closes **8:00 AM Tuesday, Nov 17** — the day before Exercise 12 and Homework 12 are due. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Python tutorial — what sensitivity analysis is for, one-at-a-time (OAT) perturbation, tornado diagrams, and how a fitted coefficient gives you a sensitivity for free. Several questions use the bid-price model from Week 11. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. Sensitivity analysis in engineering models is primarily used to:

- A) Determine whether the model is statistically significant
- B) Quantify how much the model output changes when one or more inputs change
- C) Replace regression with a simpler calculation method
- D) Eliminate uncertainty in the input data entirely before the model is ever run

Q2. [True/False] For a linear model $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$, the sensitivity of $\hat{y}$ to x_1 is just β_1 , and it is the same at every value of x_1 .

Q3. [Select all that apply] Which of the following are true of a one-at-a-time (OAT) sensitivity analysis?

- (A) It varies one input while holding all the others at their baseline values
- (B) It captures interactions between inputs
- (C) It can only perturb quantities that appear in the model
- (D) It requires the inputs to be Normally distributed

- (E) It produces a ranking of inputs by how much each moves the output
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Q4. The Week 11 price model for roadway excavation was: $\log_{10}(\text{UnitPrice}) = 1.929 - 0.166 \times \log_{10}(\text{Quantity}) - 0.054 \times \text{LOW}$

A job **ten times larger** therefore carries a predicted unit price that is approximately:

- A) 1.66% lower
 - B) 16.6% lower
 - C) 31.8% lower
 - D) unchanged — the coefficient applies to log price, not price
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Q5. A **tornado diagram** displays sensitivity results with:

- A) Horizontal bars sorted from the largest output swing (top) to smallest (bottom), one bar per input
 - B) A pie chart in which each slice represents the fraction of the output variance explained by one input
 - C) A line connecting minimum and maximum output values across all inputs simultaneously
 - D) A histogram of prediction errors for each input scenario
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Q6. [Ordering] Put the steps of an OAT sensitivity analysis in the order you would carry them out.

- (A) Fix every input at its baseline value and record the baseline output
 - (B) Perturb one input by a stated amount, leaving the rest at baseline
 - (C) Record the resulting change in output, restore that input, and move to the next
 - (D) Rank the inputs by the size of their swings and draw the tornado diagram
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Q7. A bid tabulation lists **every bidder's** price for every item on a job. To total the excavation quantity an agency actually awarded across a programme, you should:

- A) Sum Quantity over every row in the file, since each row is one item on one job
- B) Sum Quantity over every row, then divide by the number of districts that let the work
- C) Average Quantity across the rows for each project, then sum those project averages
- D) Sum Quantity over the rows where LowBidder is Yes

Q8. At a 2,000 CY design point the four swings in item cost are: unit price **\$59,144**, quantity **\$15,138**, overhead **\$4,465**, escalation **\$2,883**. Unit price tops the ranking because:

- A) Cost is linear in all four, so price wins on the width of its range, not on its response
- B) Item cost responds more steeply to unit price than to any of the other three inputs
- C) Unit price is the largest single number appearing anywhere in the cost equation
- D) The price model has the lowest R^2 of the four, so its predictions are least reliable

Q9. The quantity swing is proportional to the take-off tolerance, and at $\pm 15\%$ it is **\$15,138**. The overhead swing is **\$4,465** and does not depend on the take-off at all. Below roughly what take-off tolerance does quantity stop being the second-ranked input?

- A) $\pm 10\%$
- B) $\pm 4.4\%$
- C) $\pm 1\%$
- D) It never does — quantity is second at every tolerance

Q10. [Matching] Match each tool to the question it answers.

Choices:

- (A) Sensitivity analysis
- (B) Hypothesis testing
- (C) A tornado diagram
- (D) A regression coefficient
- (E) Monte Carlo simulation

Matches:

1. Is an observed difference larger than chance alone would produce?
2. The change in the outcome per unit change in one predictor, holding the others fixed
3. How much does the prediction move when an input moves?
4. Which inputs matter most, ranked by the swing each one causes

Q11. What does OAT (one-at-a-time) sensitivity mean in practice?

- A) Perturb every input at once and record the total change in the model's output

- B) Fit the model once per input and compare the resulting R^2 values against each other
 - C) Perturb one input, hold the rest at baseline, record the change, then restore it
 - D) Remove one input at a time, refit, and see how far the fit degrades without it
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Q12. In a linear model, the analytical sensitivity of the output to a predictor is which quantity?

- A) Its p-value, which says how surprising the coefficient would be under the null
 - B) Its correlation with the response, computed before any other predictor enters
 - C) Its share of the model's R^2 , measured by how far the fit drops when it is dropped
 - D) Its regression coefficient — the partial derivative of the prediction with respect to it
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Q13. How are the bars in a tornado diagram ordered, and why does that shape give it its name?

- A) By the size of the output swing, largest at the top, so the bars taper downward
 - B) Alphabetically by input name, so that a reader can find a particular one quickly
 - C) By the p-value of each input, with the most statistically significant one on top
 - D) Chronologically, in whatever order the inputs happened to be perturbed by the code
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Q14. Quantity, overhead and escalation get ranges an estimator chooses. Unit price instead gets ± 1 RMSE, taken from the model's own residuals. Why not simply use $\pm 10\%$ for price too?

- A) Because a $\pm 10\%$ range would be rejected by the tornado plotting code as invalid input
 - B) Because the residual spread measures how far the model's price can sit from a real bid, while $\pm 10\%$ is arbitrary
 - C) Because unit price is measured in dollars while the other three inputs are not
 - D) Because the RMSE always comes out wider than 10%, which makes the tornado clearer
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Q15. The interaction model fits $\log p \sim \log q * LOW$ and returns **two** predicted prices at the same 2,000 CY job — \$21.45 and \$24.07. Why two rather than one?

- A) A rounding artefact that is corrected later
- B) One prediction includes overhead and escalation while the other leaves them out
- C) Because the interaction lets the slope depend on LOW — winners and losers do not share one line
- D) One is the analytical prediction and the other is the numerical OAT check